

Offline, once: mapping & naming

phone capture → COLMAP + board alignment → 3DGS (board frame) → SAM cross-view instances → named registry: 259 instances · 15 named

Wearer gaze → world fixations

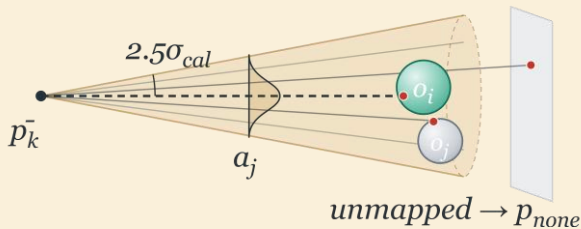
fisheye ArUco PnP
(wall tags)

undistort · bias ·
pose → world ray

cluster → fixations F_k

stable while walking

Conical posterior voting



$$a_j = \exp(-\theta_j^2 / 2\sigma_{cal}^2), \quad V_i = \sum a_j \alpha_j$$
$$q_i \propto \pi_i V_i, \quad p_{none} = 1 - \sum V_i / Z$$

$$q_i \geq \tau \rightarrow \hat{o} \quad \text{else reject } \emptyset$$

Shared 3DGS instance map one metric frame for human & robot



local depth + opacity inside the cone

persistent instance IDs shared with robot

Robot execution

LIO-SAM + A* nav
tags → world frame

move · pick · place

last-meter: hint → grasp cam

lastmeter.jpg

Speech → command

streaming ASR → LLM parse: action + slots → deictic word ↔ fixation window → confirm → $c = (a, \hat{o}, d)$